Graphs and optimisation

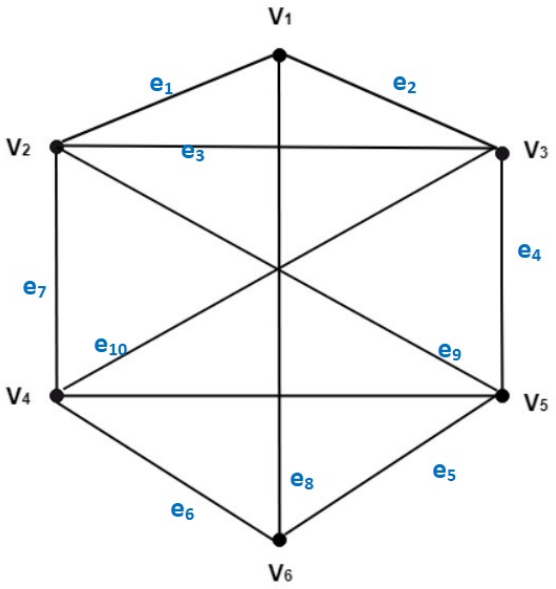
Series 2

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Exercise 1

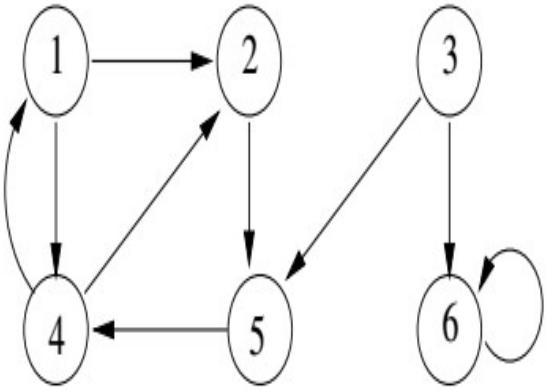
Consider the graph shown in the figure below.



1. Define the sets V and E and define the incidence function, Ψ, corresponding to graph G.
2. Give the order and size of graph G.
   Give an example of the following:
3. A simple path, of length 4, from V_1 to V_6.
4. What is the distance between V_2 and V_6?
5. What is the diameter of graph G?
6. An elementary path from V_1 to V_6.
7. A simple path which is not elementary from V_1 to V_6.
8. A path which is not simple and starting from V_2.
9. A simple circuit starting from V_1
10. A circuit which is not simple and starting from V_2.

Exercise 2

Consider the following directed graph $G = (V, E, \Psi)$:

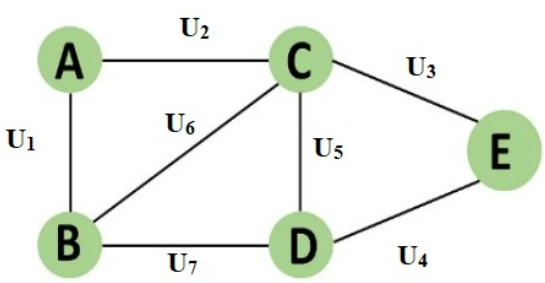


1. Define the sets V and E and define the incidence function, Ψ, corresponding to graph G.
2. Give the order and size of graph G.
3. The graph G is a p-graph. Give the value of p.
4. Give an elementary path in this graph G.
5. Give a non-elementary path in this graph G.
6. Give an elementary circuit in this graph G.
7. Give a non-elementary circuit in this graph G.
8. Is there an Eulerian path in this graph G?
9. Is there a Hamiltonian circuit in graph G?

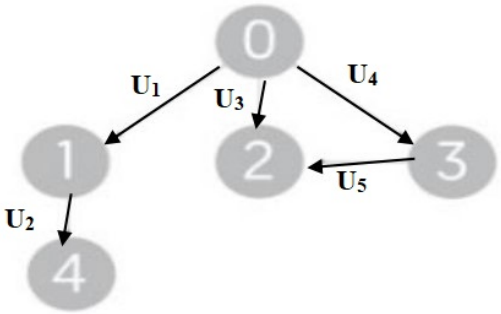
Exercise 3

1. How to detect a cycle in a directed graph using Depth First Search (DFS) algorithm?
2. How to detect a cycle in an undirected graph using DFS?
3. Why does DFS use a ”stack” data structure? Can it use a ”queue” data structure?
4. Can we use DFS to solve the shortest path distance between two nodes in unweighted graphs?

Exercise 4



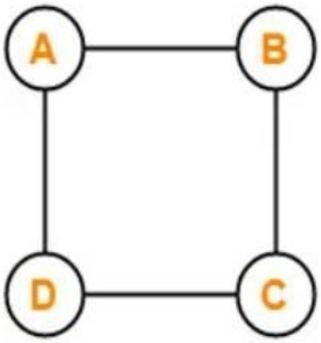
Graph G1



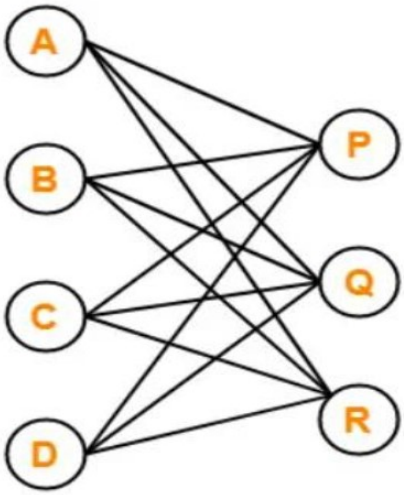
Graph G2

1. Remind the different steps to perform a depth first search (DFS) algorithm.
2. Define the sets V and E and define the incidence function, Ψ, corresponding to each graph $G_i, i = 1, 2$.
3. Dress the table of degrees corresponding to each graph $G_i, i = 1, 2$.
4. Give an elementary chain in each graph $G_i, i = 1, 2$, if it exists.
5. Give a non-elementary chain in each graph $G_i, i = 1, 2$, if it exists.
6. Give an elementary cycle in each graph $G_i, i = 1, 2$, if it exists.
7. Give a non-elementary cycle in each graph $G_i, i = 1, 2$, if it exists.
8. By applying the DFS algorithm, show that graph $G_i, i = 1, 2$ is connected (strongly connected).

Exercise 5



Graph G_1



Graph G_2

1. How can we draw graph G_1 in such a way to obtain a bipartite graph? Justify your answer.
2. Is graph G_2 bipartite? Justify your answer.
3. Is graph G_2 complete? Justify your answer.